



UNIVERSITY OF AMSTERDAM

Digital Marketing Analytics:

Recap & Overview: Pointers for the exam: Tips & Suggestions

Digital Marketing and Analytics: Additional Slides

Academic Year 2025-2026 – Semester 1 – Block 2

AMSTERDAM
BUSINESS
SCHOOL

Business

Final Exam & Grading

Assessment & Grade Components

- (1) Final Exam: 60% minimum 5,5
- (2) Team Assignments: 40% (each 20%) minimum 5,5
- (3) Mini Assignments: has to be “sufficient” to take the final exam: no % impact on final grade.

Grades Main Team Assignment 1 will be announced shortly (1-2 days) before the final-exam.

Grades Main Team Assignment 2 these will be graded along with (same time-line) with Final Exam Grading.

If you are not contacted by me so far that your mini-assignment(s) is/are insufficient – that means your mini assignments are complete/sufficient.

All grade components will be announced on CANVAS (final course grades will be approved-announced by SIS)



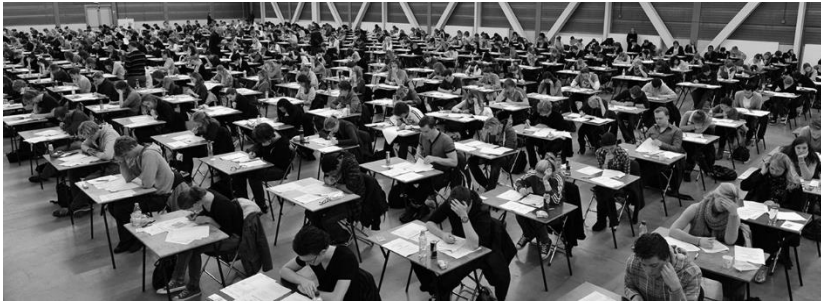
Final Exam: On-Campus Exam (Check the Exam Location Carefully on CANVAS)

- Please keep an eye / double check locations times on Rooster: uva.rooster.nl
- Friday **19 December 15:00-17:00**. Duration **2 hours**.
- There is **NO online alternative** or substitute for this exam. All students have to be at the pre-announced exam location to take the exam.
- If a student misses the final examination for any reason – the student has to take the resit exam on **02 February**.
- This is a **CLOSED BOOK exam** – so you will **NOT** be able to use your course materials during the exam. Please **prepare beforehand**.



Final Exam: On-Campus Exam (Check the Exam Location Carefully on CANVAS)

- ✓ **CLOSED BOOK exam**: NO course material during the examination.
- ✓ Both **open-ended** and **multiple-choice** questions are used in the exam. Open-Ended questions has more share 60-70 out of 100 points.
- ✓ There are 2-3 Open-Ended (you write your answers) questions ; and 5-6 Multiple-Choice questions in the exam.
- ✓ Numerical/Analytical – Verbal questions are balanced (roughly half and half)
- ✓ All questions (reflective or applied) are doable by using our course learnings = **lecture + tutorial slides**.
- ✓ The time is more than enough if you study-prepare in advance. Thus; you will not be time-pressured.
- ✓ A SAMPLE exam reflecting this year's exam structure will be available on CANVAS.



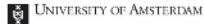
Final Exam: Course Material Required

Lecture Slides: All weeks: your **main preparation base**: obligatory for exam – to have a good summary: see following slides

Tutorial Slides: IF I have slides for that week's tutorial (Week 1 & Week 2)

All Articles: Obligatory for the exam (main findings – **summarized on slides**: no need to know the details- know the essence !)

*You are responsible for the 'essence': **objectives, conceptual framework, main findings and managerial implications** of articles and **NOT for everything in the article**; slides and student presentations give a good summary of the articles.*



Digital Marketing Analytics:
Introduction, Online-Data & Metrics, Conversion-Attribution I

Digital Marketing and Analytics: Week 1
Dr. Umut Konus

29 October 2020



Digital Marketing & Analytics - Literature List (2020-2021)

Week 1: Introduction: Introduction to Digital Marketing Analytics and Metrics, Conversion Attribution I

- Article 1.1: Matz, S. C., & Nyffeler, O. (2017). Using big data as a window into consumers' psychology. *Current Opinion in Behavioral Sciences*, 18, 7-12.
- Article 1.2: Pabjolo, H., Preiser K. and Christensen, T., (2012) Using Google Analytics to measure visitor statistics: The case of food composition websites. *International Journal of Information Management*, (32) 504-512
- Article 1.3: Nisar T. M and Young M. (2017). Attribution Modeling in Digital Advertising: An Empirical Investigation of the Impact of Digital Sales Channels. *Journal of Advertising Research*, (November)

Week 2: Conversion-Attribution II / Customer Journeys on Digital Platforms & Cross Channel Management

- Article 2.1: De Jager, L., Kannan, P.K., Verhoef P.C., and Wiesel, I. (2018) Device Switching in Online Purchasing: Examining the Strategic Contingencies. (82), 1-19
- Article 2.2: Li, J., Abbasi, A., Cheema, A., & Abraham, L. B. (2020). Path to Purpose? How Online Customer Journeys Differ for Hedonic versus Utilitarian Purchases. *Journal of Marketing*, 002247520911638.
- Article 2.3: Demmers, J., Willemsen, J. W., & van Dolen, W. M. (2020). Consumer Engagement with Brand Posts on Social Media in Consecutive Stages of the Customer Journey. *International Journal of Electronic Commerce*, 24(1), 53-77.
- Article 2.4: Hoock, L., & Spann, M. (2020). An experimental analysis of the effectiveness of Multi-Screen advertising. *Journal of Interactive Marketing*, 50, 81-99.



Tutorial 1: Introduction for Analyzing Online Data & Regression Models (Conversion-Attribution)

Digital Marketing and Analytics: Week 1
Dr. Umut Konus

29-30 October 2020



Final Exam: General Suggestions – Lecture & Tutorial (W1-W2) Slides

- ✓ Please go through/study all the slides at least one preferably more times carefully – by taking notes - before the exam.

Slides will provide you with:

- ✓ Main content of this course & Good summary of what we need to learn (with key topics, findings & examples)
- ✓ Also includes main findings/content from the articles (our reading list)
- ✓ Slides 13-19 further in this slide set will give you a concrete pointers on which topics-themes are relevant-valuable as pre-exam learnings.

Please take your time to prepare (slides, articles, tutorial slides) in advance !



UNIVERSITY OF AMSTERDAM

Digital Marketing Analytics:
Introduction, Online-Data & Metrics, Conversion-Attribution

Digital Marketing and Analytics: Week 1
Dr. Unut Kimes

28 October 2023



Digital Marketing & Analytics – Literature List (2020-2021)

Week 1: Introduction: Introduction to Digital Marketing Analytics and Metrics, Conversion Attribution

- Article 1.1: Mats, S. C., & Janssen, G. (2017). Using Big data as a window into consumers' psychology. *Current Opinion in Behavioral Sciences*, 18, 7-12.
- Article 1.2: Fildes, H., Preiser, K., and Christensen, T. (2012). Using Google Analytics to measure visitor statistics: The case of food composition websites. *International Journal of Information Management*, 32(5)504-512
- Article 1.3: Nasar, T. M. and Young, M. (2012). Attribution Modeling in Digital Advertising: An Empirical Investigation of the Impact of Digital Sales Channels. *Journal of Advertising Research*, (November)

Week 2: Conversion Attribution II / Customer Journeys in Digital Marketing & Cross-Channel Management

- Article 2.1: De Jong, E., Karmali, P.K., Voorhees P.C., and Wiesel, T. (2018) Device Switching in Online Purchasing: Implications for Strategic Contingencies. *IS*, 3-19
- Article 2.2: Li, L., Mittal, A., Chinnai, A., & Abraham, L. B. (2020). Ready to Purchase? How Online Customer Journeys Differ for omnichannel Versus Multichannel Purchases. *Journal of Marketing*, 0022420920911628.
- Article 2.3: Decker, J., Wittenwever, L. W., & van Dolyn, W. M. (2019). Consumer Engagement with Brand Posts on Social Media in Consumer Stages of the Customer Journey. *International Journal of Electronic Commerce*, 24(3), 50-79.
- Article 2.4: Haggis, L., & Sparr, M. (2008). An experimental analysis of the effectiveness of Multi-Screen advertising. *Journal of Interactive Marketing*, 22, 81-95.



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Tutorial 1: Introduction for Analyzing Online Data & Regression Models (Conversion-Attribution)

Digital Marketing and Analytics: Week 1
Dr. Unut Kimes

28-08-October 2023



Final Exam: General Suggestions – Articles

You are responsible for **all the articles** in the reading list in the exam! No matter they were presented or not

Objective: Learn how to use an article in an efficient and practical way. Focus **on the essence** not in small/methodological details

- Main findings, conclusions
- Main motivation-research question
- Main conceptual framework
- Managerial Implications

Good News: For most (almost all) articles a summary of main findings are **already given in the lecture slides/ student presentations**

Digital Marketing & Analytics - Literature List (2020-2021)

Week 1: Introduction: Introduction to Digital Marketing Analytics and Metrics, Conversion-Attribution I

- Article 1.1: Matz, S. C., & Netzer, O. (2017). *Using big data as a window into consumers' psychology*. Current Opinion in Behavioral Sciences, 18, 7-12.
- Article 1.2: Pakkias, H., Presser K. and Christensen, T., (2012) *Using Google Analytics to measure visitor statistics: The case of food composition websites*, International Journal of Information Management, (32) 504-512
- Article 1.3: Nisar T. M and Yeung M. (2017). *Attribution Modeling in Digital Advertising: An Empirical Investigation of the Impact of Digital Sales Channels*, Journal of Advertising Research, (November)

Week 2: Conversion-Attribution II / Customer Journeys on Digital Platforms & Cross Channel Management

- Article 2.1: De Haan, E., Karthian, P.K., Verhoef P.C., and Wiesel, T. (2018) *Device Switching in Online Purchasing: Examining the Strategic Contingencies*, (82), 1-19
- Article 2.2: Li, J., Abbasi, A., Cheema, A., & Abraham, L. B. (2020). *Path to Purpose? How Online Customer Journeys Differ for Hedonic Versus Utilitarian Purchases*. Journal of Marketing, 0022242920911628.
- Article 2.3: Demmers, J., Weltewreden, J. W., & van Dolen, W. M. (2020). *Consumer Engagement with Brand Posts on Social Media in Consecutive Stages of the Customer Journey*. International Journal of Electronic Commerce, 24(1), 53-77.
- Article 2.4: Hooeck, L., & Spann, M. (2020). *An experimental analysis of the effectiveness of Multi-Screen advertising*. Journal of Interactive Marketing, 50, 81-99.

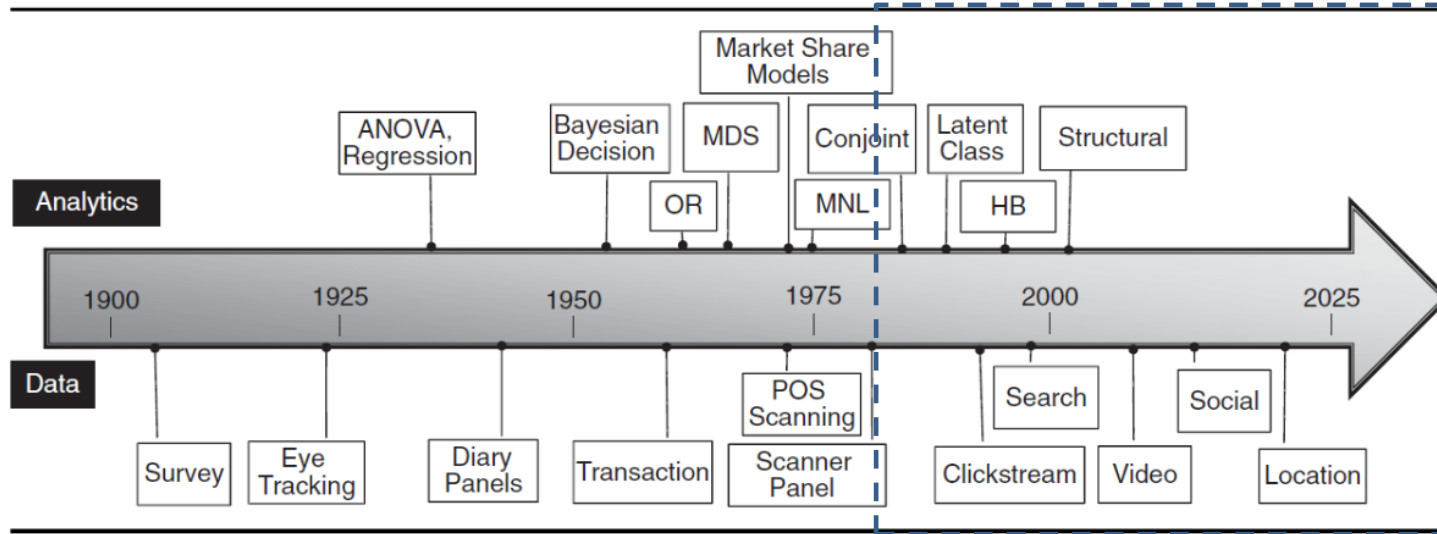
DMA: An Overview

What is There & What did we cover

Digital Marketing Analytics: Overview & Timeline

Here is an evolutionary timeline of Digital Marketing & Analytics: on top (i) methods and below (ii) data types – over the years.

In this course we focused on most novel/new data types & approaches you see on bottom-right.

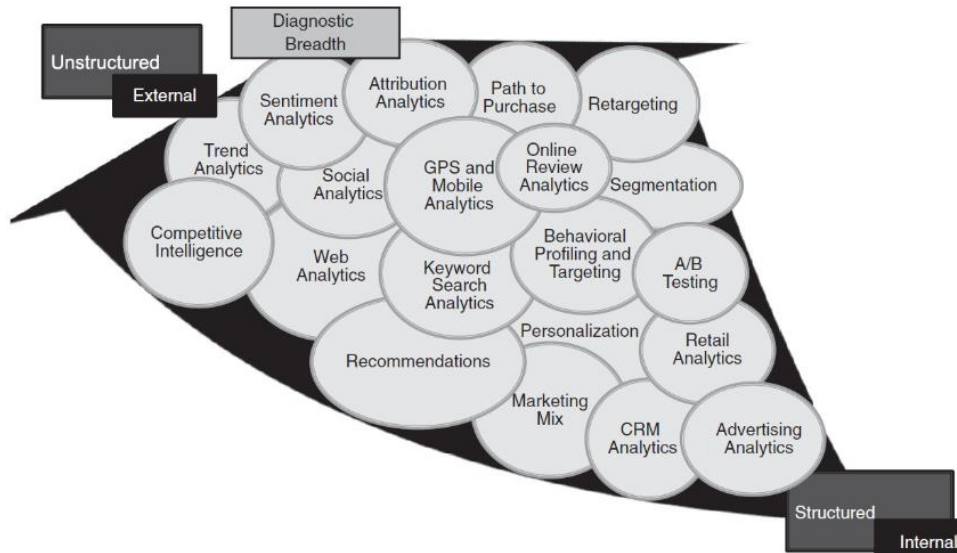


Digital Marketing Analytics: Overview & Timeline

These are the current novel & valuable topics in Digital Marketing & Analytics: each of them well deserves a separate course for marketing analysts.

This course takes a holistic perspective and **introduce you with most of them** in brief: **(what, when, what for, how ?)**

Now it is up to you to go deeper and gain expertise on the ones that you are interested & that you find more valuable for your career objectives.



Key Topics :
What to Know for (and before) the Exam

Key Topics (or) What Do I Need to Know? – Week 1

- Paid, Owned, Earned, Category Touchpoints (Article 1.3) Media, Key KPI's-Metrics used in Digital Marketing
- Digital Data Types: On-site, Clickstream and Online CRM – Structured vs Unstructured – Zero, First, Second and Third Party
- Concept of Journey (i.e. Article 1.1) Main Attribution methods/techniques w/ regard to different media types (including slides + overview in Article 1.2)
- How (simple) attribution techniques work? (Last-Click, First-Click, Linear, Position-Based and Time-Decay) – Calculating Attribution value
- Model Based Attribution: Logistic Regression (interpreting results: Tutorial 1)
- Probabilistic Attribution: Calculating touchpoint attribution-value in a journey depending on its position-sequence in journey.
- Modeling/Analyzing Online-Data (Sales, Review Scores etc.): Multiple Linear Regression: Tutorial 1 – dummies and concept of interaction*



Digital Marketing Analytics:
Introduction, Online-Data & Metrics, Conversion-Attribution I

Digital Marketing and Analytics: Week 1
Dr. Umut Korus

26 October 2020



Tutorial 1: Introduction for Analyzing Online Data & Regression Models (Conversion-Attribution)

Digital Marketing and Analytics: Week 1
Dr. Umut Korus

26-30 October 2020



* We will not ask you to run/analyze the data during the exam. However, we can ask you to statistically interpret/draw solid conclusions from analysis results (regression or attribution model outputs)

Key Topics (or) What Do I Need to Know? – Week 2

- Challenges in Attribution: Time Window, View-Through Attribution & Multi-Device and Multi-Platform Attribution
- Marketing/Media Mix Models – how used as an alternative to Attribution. Advantages/Disadvantages.
- Linking Web Traffic to Online Purchases (**Article 2.1**) explaining online sales by using a small set of channel exposures.
- Online, Offline and Cross-Channel Customer Journeys: Carry-Over and Spill-Over Effects, Online X Offline Interplay & Research Shopping Types
- Social Media: basic concepts - platforms, influence, influencers, virality, basic metrics & understanding social media impact/consequences
- Unstructured Data: basics – text, photo and video data, Impact of Emotions/Linguistic (Text) and Photo (Components) – **Articles 3.3 and 3.4**
- Basic understanding of how to analyze* **Scraped Data and Sentiment-Topic in Textual data** (correlation, regression) – Tutorial 2.
- Main Concepts-Understanding Behind Unstructured Data & NLP (Textual Data Analysis) – (i.e. **Article 2.2. & Article 2.3**)

* We will not ask you to scrape a data or run a NLP analysis during the exam. But we can ask you to interpret/draw conclusions from a scraped data set or results/output of a sentiment analysis

Key Topics (or) What Do I Need to Know? – Week 3

- Natural Language Processing: Sentiment & Topic Analysis Analysis: Basics (slides): definition, use-benefits, challenges and shortcomings
- **How to prepare for, make decisions and use Sentiment/Topic Analysis – Tutorial 2 (in combination with Tutorial 1)**
- Social Media Modeling: Conceptual Understanding – how to link drivers/antecedents with outcomes
- Paralinguistics & Non-Verbal Elements in text data. Emoji & Emoticon Analytics – Article 2.3 & Article 3.2
- Generative AI – definition and basic of use in Digital Marketing Analytics



Social Media Analytics: Numeric, Visual and Textual Data
Sentiment Analysis

Digital Marketing and Analytics: Week 3
Dr. Umit Korum
09 November 2020



Tutorial 2: Text Analytics – Sentiment Analysis

Digital Marketing and Analytics: Week 2
Dr. Umit Korum
05 November 2020



Key Topics (or) What Do I Need to Know? – Week 4

- Online Ratings & Reviews: components, platforms, value, structure (distributions) and challenges
- Online Ratings & Reviews: structural differences – how to see/decompose a review (or rating) screen to its components for further analysis
- Online Reviews: mere presence, arousal, review components, visuals' effect on business outcomes, firm responses to user reviews (Articles 4.1, 4.2, 4.3)
- SEO: On-Page and Off-Page Optimization, RankBrain, ranking and ranking factors, Indirect effect of SEO on SEA activity-outcomes.
- SEA: Adwords & Ranking in Paid Search results, Quality Score (components), impact on business outcomes & customer response
- SEA-SEO: main differences, platforms, uses, value and challenges, long-term impact & comparison



Online Review-Rating Analytics
Search Marketing: Search Engine Advertising & Optimization

Digital Marketing and Analytics: Week 4
Dr. Unut Kocup
16 November 2020



Key Topics (or) What Do I Need to Know? – Week 5

- Digital Advertising: pricing models (CPM/V, CPC and CPA/L) main differences, comparison, advantages and disadvantages in different situations
- Digital Advertising: calculating cost per sale for CPM, CPC and CPA. The impact of Quality Score on pricing.
- Digital Advertising: Real-Time Bidding and Second Price Auctions
- **Dynamic Pricing for Consumer Side: pros- cons- benefits and risks.**
- Personalized Targeting: Levels of personalization and types/forms of individualization, comparison of targeting methods
- Impact of Emotions during Search Engine Use (Article 5.1)



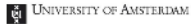
Digital Advertising: Pricing Models
Personalization & Personalized Targeting

Digital Marketing and Analytics: Week 5
Dr. Umut Konus
23 November 2020



Key Topics (or) What Do I Need to Know? – Week 6

- Recommender Systems: value, use, benefits, types: Content (Item) Based and Collaborative (User Bases) Filtering
- **Recommender Systems: comparison the impact of different framings (Article 5.2) and sales diversity (Article 5.3)**
- Mobile Marketing: key concepts, platforms, role in customer journeys, impact/consequence of device, platform, timing
- Mobile Shopping: drivers, app-use and its impact on purchases (**Article 6.1**), mindset metrics and online-offline sales
- Mobile Applications: Impact of Mobile-App Failures (**Article 6.2**) – Impact of Gamification on Engagement (**Article 6.4**)
- Mobile Applications: Platform Expansion – adding a new app in the existing portfolio & consequences (**Article 6.3**)
- Mobile Targeting: location-based targeting, value, different types-formats and comparison



Recommendation Systems
Mobile Marketing & Targeting

Digital Marketing and Analytics: Week 6
Dr. Umair Korus
30 November 2020



Final Exam: Tips & Suggestions

Final Exam: Tips and Suggestions – Open-Ended Questions

The answers are considerably short/simple (if you do the reasoning & write exactly what is expected-asked in the question)

A Typical Open-Ended Question is a question mostly focuses on:

Whether you can (i) apply your knowledge in a business case and (ii) draw necessary conclusions/implications and (iii) make decisions/come up with solutions on which approach/methods to use (with your reasons) in certain business situations:

*For example: What approach/method would you use in this (a fictional/actual business case) and why? Explain with your reasons Please – IF you are asked about reasons: briefly give your: **reasons (WHY)** are the essential part of the answer –**why/how** ?*

Important: In your reasoning/ (why part of the questions) related answer please refer to our course learnings: slides – where necessary !



Final Exam: Tips and Suggestions for Open-Ended Questions

Do not try to write-capture everything: instead **be selective and to-the-point (+)** in your answers

Reasoning (Argumentation) is very important:

Please give your concrete and specific (no generalizations) reasons-arguments behind your choices. Be consistent & congruent in your answers.

Try to base your reasons/arguments on the theory and literature (slides and articles) and business examples given in the lectures as much as possible.

For example:.... *Because It is good to increase the sales...* (without giving any solid-specific reason, would NOT be a good answer!)

There might be **1-2 little calculations** however these will be limited with (i) adding, dividing or multiplying few numbers in simple metrics, so you can easily do with pen and paper or (ii) interpreting a model/analysis results table (remember this is NOT a modeling or statistics course)



Final Exam: Suggestions on Time Management

In the exam there are 2-3 open-ended questions (with relatively short answers) + 5-7 multiple choice questions :

Thus, the time (2 hours) will be perfectly enough.



However, you may feel time pressure IF:

IF you intend to write excessive amount of things (not expected) by hoping to somehow relate with the expected answer.

The answers for open-ended questions are often short and to-the-point 2-3 sentences or few bullet points.

It will take vast amount of time to write different/ many “irrelevant” things which are not expected in the answer.



That is not what we meant by our exam > we want to assess who studied/prepared for the exam in advance:

Take your time to go through the slides and prepare – BEFORE the exam.

Be relevant, concise and to the point in your answers – while giving the reasons/argumentation for your answers (in open-ended)



Success with the Exam

Dr. Umut Konus
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